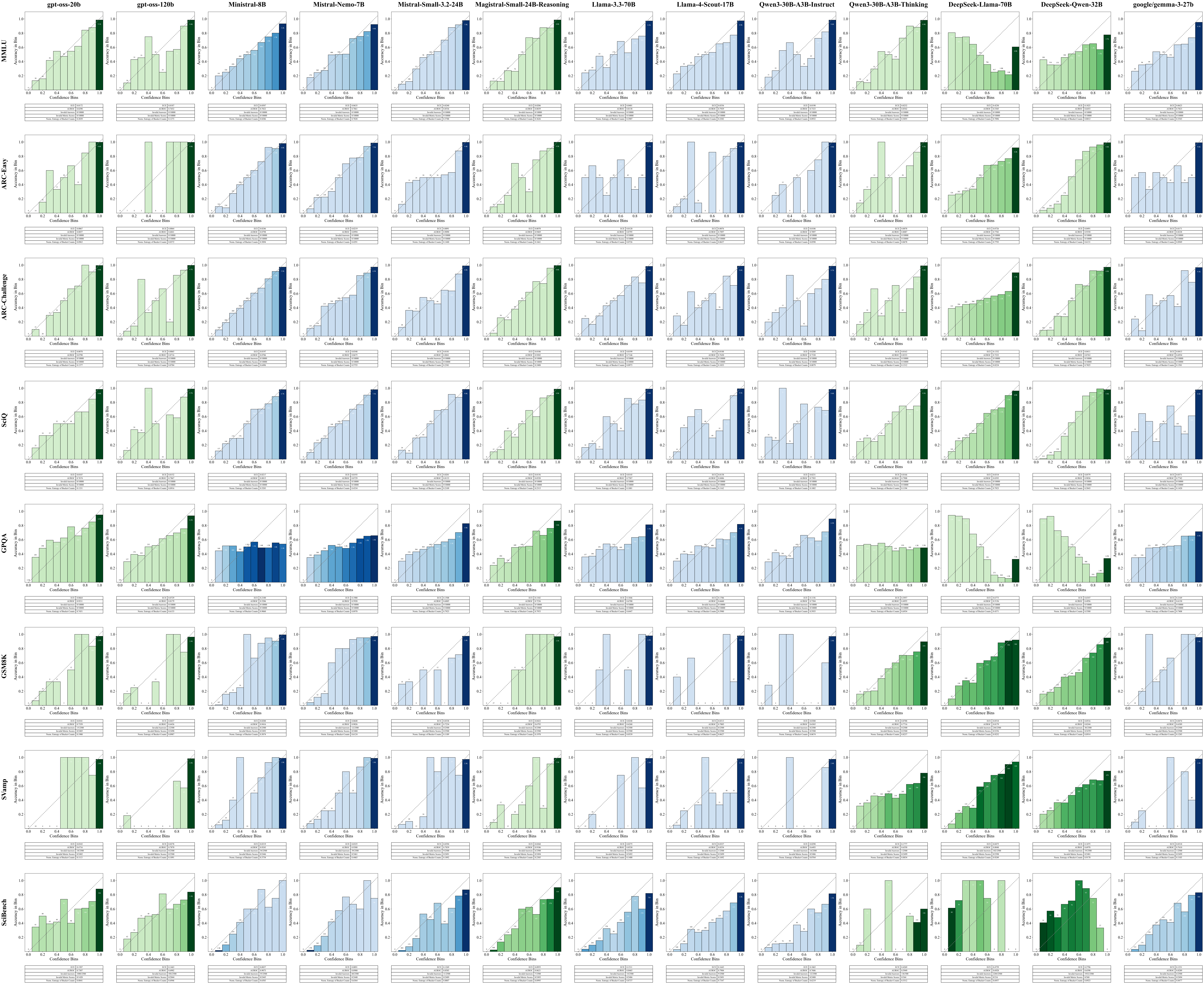
Calibration Plots for Uncertainty Metric Frequency of Answer



MMLU

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0172	0.0187	0.0587	0.0635	0.0240	0.0206	0.0401	0.0336	0.0198	0.0232	0.4230	0.1825	0.0623
AUROC	0.8290	0.7652	0.7822	0.7861	0.8536	0.8839	0.8138	0.7929	0.7323	0.8542	0.5385	0.6453	0.7823
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.2039	0.1273	0.8286	0.7828	0.3788	0.3626	0.2907	0.2282	0.0932	0.3055	0.7896	0.8012	0.3543

ARC-Easy

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0067	0.0064	0.0246	0.0219	0.0091	0.0070	0.0120	0.0076	0.0106	0.0078	0.0726	0.0491	0.0171
AUROC	0.8857	0.6841	0.8798	0.8994	0.9090	0.9469	0.8795	0.7897	0.7097	0.8087	0.7782	0.9350	0.8228
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.0963	0.0372	0.5094	0.4291	0.1440	0.1661	0.0726	0.0637	0.0598	0.0678	0.7793	0.6131	0.0949

ARC-Challenge

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0078	0.0065	0.0197	0.0348	0.0181	0.0106	0.0199	0.0188	0.0200	0.0163	0.1332	0.0411	0.0413
AUROC	0.8790	0.8716	0.8706	0.8675	0.8841	0.9202	0.7146	0.7630	0.7330	0.8333	0.7222	0.8763	0.6934
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.1377	0.0784	0.6496	0.5755	0.2381	0.2408	0.0973	0.1033	0.0879	0.1312	0.8224	0.7025	0.1501

SciQ

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0187	0.0152	0.0217	0.0193	0.0201	0.0156	0.0145	0.0138	0.0231	0.0166	0.0318	0.0579	0.0371
AUROC	0.7674	0.7867	0.8815	0.8729	0.8133	0.8520	0.7587	0.8486	0.7583	0.7908	0.8393	0.9036	0.7742
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.1331	0.0916	0.5283	0.4334	0.2109	0.2315	0.1001	0.1162	0.1082	0.1194	0.7923	0.5843	0.1420

GPQA

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0664	0.0729	0.2100	0.1900	0.1508	0.1102	0.1846	0.1706	0.1156	0.3957	0.6773	0.6547	0.2169
AUROC	0.9211	0.8179	0.5206	0.5928	0.6885	0.7236	0.6700	0.6660	0.7384	0.4955	0.5156	0.4956	0.6130
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.7451	0.6153	0.9060	0.8978	0.7923	0.7953	0.6447	0.5980	0.5955	0.6934	0.4371	0.5580	0.7408

GSM8K

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0341	0.0437	0.0288	0.0640	0.0338	0.0422	0.0248	0.0313	0.0380	0.0788	0.0518	0.0516	0.0474
AUROC	0.7359	0.6436	0.9416	0.9026	0.7234	0.6793	0.6268	0.7005	0.6565	0.7716	0.8179	0.8244	0.6389
Invalid Answers	15/2500	2/2500	8/2500	12/2500	0/2500	0/2500	0/2500	0/2500	0/2500	0/2500	244/2500	30/2500	0/2500
Invalid Metric Scores	0/2485	0/2498	0/2492	0/2488	0/2500	0/2500	0/2500	0/2500	0/2500	0/2500	0/2256	0/2470	0/2500
Norm. Entropy of Bucket Counts	0.1900	0.0987	0.5074	0.6124	0.1189	0.1970	0.0539	0.0627	0.0674	0.8237	0.9252	0.8914	0.1285

SVamp

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0343	0.0278	0.0319	0.0325	0.0394	0.0268	0.0275	0.0327	0.0298	0.1777	0.0475	0.1075	0.0318
AUROC	0.6716	0.7870	0.9243	0.9380	0.7839	0.9186	0.8530	0.8554	0.6493	0.6550	0.8048	0.6978	0.7610
Invalid Answers	18/2500	0/2500	54/2500	39/2500	9/2500	6/2500	8/2500	1/2500	14/2500	1/2500	102/2500	19/2500	1/2500
Invalid Metric Scores	0/2482	0/2500	0/2446	0/2461	0/2491	0/2494	0/2492	0/2499	0/2486	0/2499	0/2398	0/2481	0/2499
Norm. Entropy of Bucket Counts	0.1113	0.1091	0.3754	0.4465	0.1892	0.2385	0.1400	0.1692	0.0764	0.8834	0.9249	0.9170	0.1103

SciBench

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.1587	0.1463	0.0837	0.0891	0.1269	0.0955	0.1511	0.1643	0.1865	0.4269	0.4759	0.2796	0.1231
AUROC	0.7307	0.6902	0.9073	0.8980	0.8585	0.8621	0.8465	0.7984	0.7686	0.5949	0.6928	0.6598	0.8289
Invalid Answers	1080/2500	456/2500	55/2500	60/2500	11/2500	7/2500	14/2500	9/2500	12/2500	136/500	2284/2500	1932/2500	6/2500
Invalid Metric Scores	0/1420	0/2044	0/2445	0/2440	0/2489	0/2493	0/2486	0/2491	0/2488	0/364	0/216	0/568	0/2494
Norm. Entropy of Bucket Counts	0.8041	0.6946	0.6583	0.6564	0.8807	0.8995	0.8737	0.7347	0.6219	0.5512	0.6957	0.8923	0.8477